

Sen Wang

Ph.D. Candidate in Artificial Intelligence and Robotics, IAIR, Xi'an Jiaotong University

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Google Scholar | GitHub

Research Interests

Robotic manipulation; Generative World Models; Latent World Action Model; 3D Scene Representations.

Education

Xi'an Jiaotong University

Ph.D. Candidate in Artificial Intelligence and Robotics

Xi'an, China

Sept. 2023 – Present

- Advisor: Prof. Le Wang

Jilin University

B.Eng. in Control Science and Engineering

Jilin, China

Sept. 2019 – Jun. 2023

- GPA: 3.8/4.0; undergraduate advisor: Prof. Yan Ma

Publications and Manuscripts

- [1] **S. Wang**, L. Wang, H. Huo, S. Zhou, K. Xia, and G. Hua. “Integrating Agglomerative Perception with One-step Action Generation for Robotic Manipulation.” *TPAMI*, 2026.
- [2] **S. Wang**, H. Dong, J. Tian, J. Li, T. Cao, A. Chen, S. Wu, S. Zhou, and L. Wang. “Efficient Camera Pose Augmentation for Viewpoint Generalization in Robotic Policy Learning.” *ECCV*, 2026.
- [3] **S. Wang**, L. Wang, S. Zhou, J. Tian, J. Li, H. Dong, G. Hua, and W. Tang. “SAMPO: Scale-wise Autoregression with Motion Prompt for Generative World Models.” *NeurIPS*, 2025.
- [4] **S. Wang**, L. Wang, S. Zhou, J. Tian, J. Li, H. Sun, and W. Tang. “Grounding Flow Matching Policy with Region-Aware Mamba Framework for Robotic Manipulation.” *CVPR*, 2025.
- [5] Z. Yang, S. Zhou, **S. Wang**, J. Tian, J. Li, H. Gang, and L. Wang. “RAGrasp: A Retrieval-Augmented Framework with Diversity-Aware Modeling for Dexterous Grasp Generation.” *ECCV*, 2026.
- [6] J. Tian, L. Wang, S. Zhou, **S. Wang**, J. Li, and G. Hua. “DynaRend: Learning 3D Dynamics via Masked Future Rendering for Robotic Manipulation.” *NeurIPS*, 2025.
- [7] J. Tian, L. Wang, S. Zhou, **S. Wang**, J. Li, H. Sun, and W. Tang. “PDFactor: Learning Tri-Perspective View Policy Diffusion Field for Multi-Task Robotic Manipulation.” *CVPR*, 2025.
- [8] **S. Wang**, S. Zhou, H. Dong, K. Xia, G. Hua, and L. Wang. “Unified Temporal Autoregressive Modeling with Scale-Wise Flow Matching for Action-Conditioned World Models.” *TPAMI*, *under review*.
- [9] **S. Wang**, S. Zhou, H. Dong, K. Xia, G. Hua, and L. Wang. “ReLaT-E: End-to-End Relational Latent Tuning for Multimodal Video Generation.” *under review*.
- [10] **S. Wang**, S. Zhou, H. Dong, K. Xia, G. Hua, and L. Wang. “TriFlow: Scalable Video Generation with Latent Triplane Flow Matching.” *under review*.
- [11] H. Dong, S. Zhou, **S. Wang**, J. Tian, K. Xia, and L. Wang. “RAPA: Reason-Aware Visual Policy with Preference Alignment for Zero-shot Navigation.” *under review*.

Research Experience

Horizon Robotic Lab

Research Intern

Apr. 2026 – Present

- Conducting research on **Latent World Action Model**, with an emphasis on learning compact latent dynamics for action-conditioned world modeling and robotic policy learning.

Huawei Noah's Ark Lab

Research Intern

May 2025 – Jan. 2026

- Developed **GenSplat**, a feed-forward 3D Gaussian Splatting framework for geometrically consistent novel-view synthesis from sparse, uncalibrated images.
- Applied synthesized camera viewpoints to improve viewpoint generalization and robustness in robotic policy learning.

Honors and Awards

Academic Scholarship , Xi'an Jiaotong University	<i>2024, 2025</i>
Runner-up , 5th Embodied AI Workshop (EAI)	<i>2024</i>
Freshman Scholarship , Xi'an Jiaotong University	<i>2023</i>
Outstanding Graduate Student , Jilin University	<i>2023</i>
National Encouragement Scholarship , Ministry of Education of China	<i>2021, 2022</i>

Patents

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- L. Wang, **S. Wang**, S. Zhou, and K. Xia. "A robot action generation method and related apparatus." Chinese Patent ZL 2025 1 0284999.5, issued Nov. 2025.

Technical Skills

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- **Programming:** Python, PyTorch, C, MATLAB
 - **Research Methods:** robotic policy learning, 3D Gaussian Splatting, flow matching, generative world modeling, video generation
 - **Languages:** Mandarin Chinese; English (CET-6: 536)